Phylogenetic Reconstruction by Maximum Parsimony

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1 Introduction

Phylogenetic reconstruction is a fundamental task in evolutionary biology, aiming to infer the evolutionary relationships among a set of species or genes. In this work we discuss the maximum parsimony (MP) method for phylogenetic reconstruction. It builds a phylogenetic tree based on a taxa alignment by minimizing the total number of evolutionary changes between successive nodes in the tree. This method is based on the principle of parsimony, which states that the simplest explanation (i.e., the one with the fewest changes) is preferred.

2 The Maximum Parsimony Method and the Fitch-Hartigan Algorithm

2.1 Maximum Parsimony Method

The *Maximum Parsimony (MP)* method, often referred to simply as parsimony, is designed to address the problem of *phylogenetic tree inference*. This problem consists in determining, given some observed data as input–such as DNA sequence alignments from multiple taxa–the phylogenetic tree that best explains the data.

The MP method is one possible approach to this problem. Rather than relying on an explicit probabilistic model, it seeks the tree that minimizes the total number of evolutionary changes required to account for the observed data.

To introduce this method, let us first formalize the theoretical framework in which this problem is studied.

2.1.1 Sequence Data and Characters

The MP method can be applied to phylogenetic tree inference using different types of data, such as sequence data or other forms of information, including morphological data. In this work, we focus on sequence data, specifically on DNA sequence alignments, as an illustrative example.

We assume as input a DNA sequence alignment from multiple taxa, that is, a collection of sequences composed of the nucleotides adenine (A), thymine (T), guanine (G), and cytosine (C). Each sequence corresponds to a different taxon; we therefore refer to it as *taxa data*, and denote it by the symbol X.

Therefore, our input will be a *taxa data set*, denoted by

$$\mathbf{X} = \{X_1, \ldots, X_n\},$$

where each X_i is a sequence of length k with characters in the set of nucleotides $\Sigma = \{A, G, T, C\}$. An example of this type of data is shown in Figure 2.

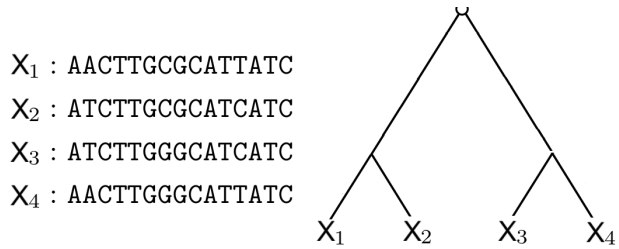

Figure 1: Example of taxa data and phylogenetic **X**-tree.

Given the set of taxa data **X**, the goal of phylogenetic tree inference is to find a phylogenetic tree T that best explains the data. That is, we assume that each taxa data X_i is generated at a distinct leaf of an underlying phylogenetic tree, and we seek to infer this tree from the data observed at the leaves.

We refer to such a tree as a *phylogenetic* **X**-*tree* T, and we abuse notation by identifying the set of leaves with the taxa data set, that is, $L(T) = \mathbf{X}$.

Let us now formalize the concept of a character with respect to the data.

Definition . A *character* with state set S on a set of taxa $\boldsymbol{X}$ is a function

$$\chi : \boldsymbol{X} \to S.$$

For DNA k-sequences, we consider $S = \Sigma$, and the set of characters

$$C = \{\chi_1, \ldots, \chi_k\}, \quad \chi_i \equiv i\text{th site in the } k\text{-sequence}.$$

This means that, given any sequence of nucleotides, the character χ_i maps the sequence to the nucleotide at the ith position.

2.1.2 Parsimony Score and Parsimonious Trees

In this theoretical framework, we can introduce the simplest form of the principal underlying parsimony:

The best tree to infer from data is the one requiring the fewest changes in states of characters.

To formalize this principle, we first note that, to determine the number of state changes required by T, we need to consider characters on the full vertex set $V(T)$, not just at the leaves. For this reason, we introduce the following definition:

Definition . An *extension of a character* $\chi : L(T) \to S$ *to* T is a character

$$\tilde{\chi} : V(T) \to S \quad \text{such that} \quad \tilde{\chi} \restriction_{L(T)} = \chi.$$

For DNA k-sequences, we consider the set of character extensions $\tilde{C} = \{\tilde{\chi_1}, \ldots, \tilde{\chi_k}\}$.

We also denote by $Ext_T(\chi)$ the set of all extensions of χ to T, and we may interpret its elements $\tilde{\chi}$ as representing the possible evolutionary histories that are consistent with the observed character χ.

Since we want to count the number of state changes of characters, we may formalize a state change between two vertices of the tree:

Definition . For an edge $e = \{v, w\} \in E(T)$ and a character extension $\tilde{\chi}$ on $V(T)$, we define the *state-change indicator* as the function

$$\delta(e, \tilde{\chi}) = \begin{cases} 1 & \text{if } \tilde{\chi}(v) \neq \tilde{\chi}(w), \\ 0 & \text{otherwise.} \end{cases}$$

Viewed as a function of e, with $\tilde{\chi}$ fixed, this is the indicator function of the edges where a state change occurs in the evolutionary history $\tilde{\chi}$. The total number of state changes in the tree can then be counted as the sum of the state-change indicators over all edges of the tree:

Definition . The *state-change count* for $\tilde{\chi}$ is the number of edges on which a state change occurs for $\tilde{\chi}$:

$$c(\tilde{\chi}, T) = \sum_{e \in E(T)} \delta(e, \tilde{\chi}).$$

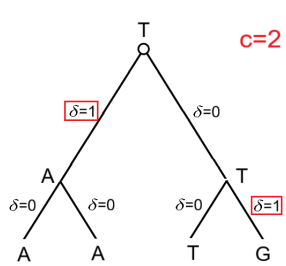

Figure 2: Example of the state-change count for an extension of a character.

Now that we have formalized the counting of state changes in the tree, we can introduce the fundamental concept for MP:

Definition . The *parsimony score* for a character χ is the minimal state-change count achieved by an extension of χ:

$$ps_\chi(T) = \min_{\tilde{\chi} \in \mathrm{Ext}_T(\chi)} c(\tilde{\chi}, T).$$

We say that $\tilde{\chi}$ is a *minimal extension* of χ for T if

$$c(\tilde{\chi}, T) = ps_\chi(T).$$

We note that these definitions are for a single character, but our data consists of the set of k characters C. We may extend the previous definition to this case:

Definition . The *parsimony score* of a phylogenetic $\mathbf{X}$-tree T for a set of characters $\{\chi_1, \ldots, \chi_k\}$ is the sum of the scores for the individual characters:

$$ps_{\{\chi_i\}}(T) = \sum_{i=1}^{k} ps_{\chi_i}(T).$$

Finally, the goal of the MP method is to determine the following set:

Definition . The set of *most parsimonious trees* for a set of characters $\{\chi_1, \ldots, \chi_k\}$ is the collection of trees achieving the minimal parsimony score:

$$\{\, T \mid ps_{\{\chi_i\}}(T) = \min_{T'} ps_{\{\chi_i\}}(T') \,\}.$$

2.1.3 Small and Large Parsimony Problems

The goal of the Maximum Parsimony method is to find the set of most parsimonious trees for a given taxa data set $\mathbf{X}$. We note that two different optimization problems can be distinguished in this method:

1. Computing the parsimony score for a fixed tree, which is known as the *small parsimony problem.*
2. Computing the parsimony score over all trees, which is known as the *large parsimony problem.*

On the one hand, the small parsimony problem can be solved efficiently using the Fitch-Hartigan algorithm, which we introduce in the following section.

On the other hand, the large parsimony problem is a difficult problem, since it involves searching over a very large set of trees, as we discuss in Section 5. Nevertheless, it is sufficient to search among the *rooted binary phylogenetic trees*, as we comment next.

First, we note that, given a phylogenetic **X**-tree, we can place a root at any location in the tree without changing its parsimony score. More precisely, we have the following result:

Lemma. If T^ρ is a rooted version of T, then for any character χ,

$$ps_\chi(T^\rho) = ps_\chi(T).$$

Proof. If ρ is a vertex of the tree T, then by the definition of the parsimony score this is clear. Now, let us assume that the root ρ has been introduced along an edge of T to obtain T^ρ, by replacing that edge with two edges meeting at ρ. Then, any minimal extension on T can be extended to a minimal extension on T^ρ without altering the state-change count, by assigning to ρ the same state as at least one of the adjacent vertices. ■

Second, we can assume that the rooted phylogenetic trees are binary, since for any rooted phylogenetic tree, there always exists an equivalent rooted binary phylogenetic tree. Figure 3 provides insight into why this is possible.

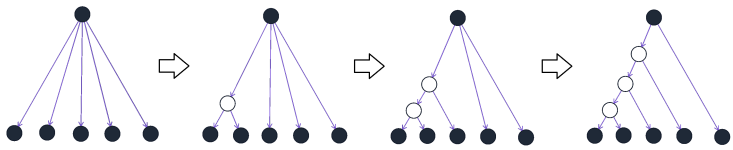

Figure 3: Schematic illustration of the recursion followed to find equivalent rooted binary trees. The illustrated process is applied to each vertex of the original tree with more than two children. The vertices with empty interiors are the interior nodes added during the recursion, and each edge whose child is one of these nodes has the identity matrix as its associated transition matrix. The transition matrices for the edges whose children are the original vertices are preserved during the recursion.

2.2 Fitch-Hartigan algorithm

2.2.1 The algorithm

Let us suppose that we want to compute the parsimony score on a given tree T, that is, we consider the small parsimony problem. One way to do this would be to proceed according to the definitions, consider all possible extensions, compute their state-change count, and take the minimum as the parsimony score. However, that approach would be very inefficient.

A simpler and more effective way was proposed independently by Fitch and Hartigan, and the resulting algorithm is named after them. For simplicity, we first introduce the algorithm for a single character. The generalization is then very straightforward.

Let T be a given binary tree, whose parsimony score we want to compute. If T is unrooted, we arbitrarily introduce a root. In the first step, we assign to each leaf X_i the corresponding state $\chi(X_i)$, that is, in our case, the nucleotide. The initial score is zero. Then we proceed from bottom to top and assign states (nucleotides) to all the internal vertices of the tree as follows. For every internal vertex, we look at the states assigned to its children. If the intersection of sets of states assigned to children is non-empty, then that intersection is assigned to the parent. If the intersection is an empty set, then the union of the sets of states is assigned to the parent. For every time the union is assigned, the score is increased by one. The process is repeated until the root is assigned and the final value of the score is obtained. A more formal description of the process is presented in Algorithm 1.

Algorithm 1 Fitch-Hartigan for a single character

Input: tree T

1. If T is unrooted, arbitrarily introduce a root ρ.
2. Assign to each leaf $X_i \in \mathbf{X}$ the pair $(\{\chi(X_i)\}, 0)$.
3. Let v be an internal vertex, and children of v were assigned pairs (U_1, m_1) and (U_2, m_2). Assign to v a pair (U, m) as follows:

   if $U_1 \cap U_2 = \emptyset$ **then**
   $(U,\, m) = (U_1 \cup U_2,\, m_1 + m_2 + 1)$
   else
   $(U,\, m) = (U_1 \cap U_2,\, m_1 + m_2)$

   Repeat for all the internal vertices.
4. If the pair $(U_\rho,\, m_\rho)$ was assigned to the root ρ, then m_ρ is the parsimony score of T.

Output: parsimony score, possible states of the characters on the internal vertices

The generalization of the Fitch-Hartigan algorithm to compute the parsimony score of a tree on sequences of more than one character is very simple. Since the parsimony score of a sequence is just the sum of scores of each character, we can proceed in the same way as described, treating the characters in

parallel.

2.2.2 Example

We consider four taxa s_1, s_2, s_3, s_4,

$$\begin{aligned} s_1 &: \quad TGCTAG, \\ s_2 &: \quad AAGTAT, \\ s_3 &: \quad TACTCG, \\ s_4 &: \quad AGGTAG, \end{aligned}$$

on a tree $((s_1, s_2), (s_3, s_4))$. Our goal is to compute the output of the Fitch-Hartigan algorithm for these sequences. Let us now consider the vertex with children s_1, s_2. We proceed character by character. If the sets of children's states have non-empty intersection (positions 4 and 5) we assign this intersection. Otherwise (positions 1, 2, 3, and 6) we assign the union. Since we have assigned the union of the sets four times, we increase the score by four. That is, we sum the children's counts (in this case both zero) and add four. We repeat this procedure for all internal vertices, as shown in Figure 4. Hence, for the given tree, we obtain the parsimony score $ps_\chi(T) = 8$.

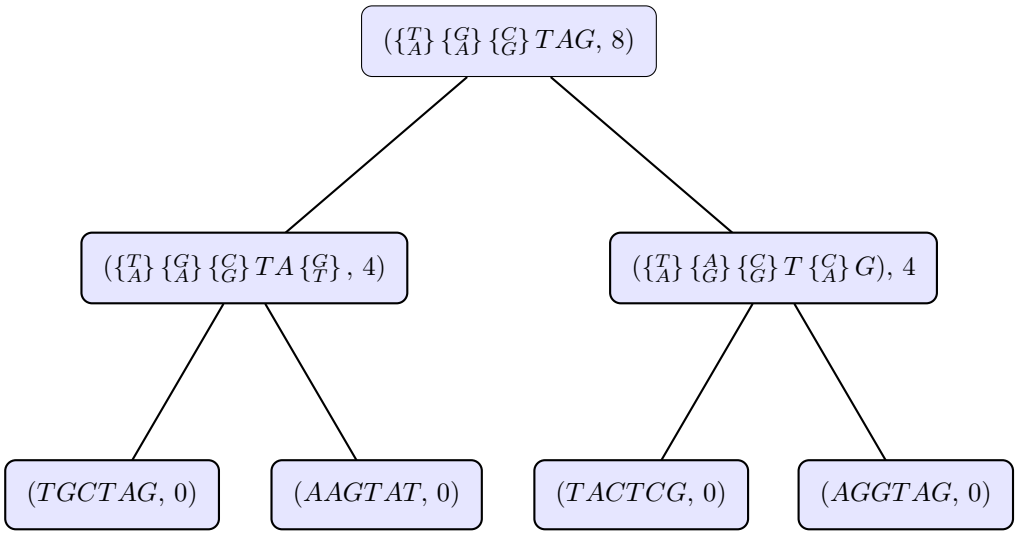

Figure 4: Computation of the parsimony score using Fitch-Hartigan algorithm.

2.2.3 Proof of correctness

Although intuitively it seems that the algorithm could work as expected and provide the true value of the parsimony score, it has to be proven rigorously.

This was done by Hartigan in 1973. We will prove the following statement.

Theorem. Let χ be a character on a finite set $\mathbf{X}$, and let T_ρ be a rooted binary phylogenetic $\mathbf{X}$-tree. Then the Fitch-Hartigan algorithm computes the parsimony score $ps_\chi(T)$.

Proof. We proceed by induction on $|\mathbf{X}|$. We start with the initial case $|\mathbf{X}| = 2$. Then the tree consists of a single edge. The Fitch–Hartigan algorithm assigns a score of 0 if the two leaf states agree and 1 otherwise, which is clearly the minimum number of changes required. Thus, the claim holds.

For the inductive step, assume that the claim holds for all rooted binary trees whose leaf sets have a size smaller than $\mathbf{X}$, and let T_ρ be a rooted binary $\mathbf{X}$-tree. Denote v_1, v_2 the children of the root ρ, and let T_1 and T_2 be the subtrees rooted in v_1 and v_2. Denote by χ_i the restriction of χ to the set of leaves of T_i. By the inductive hypothesis, the Fitch–Hartigan algorithm computes the correct parsimony scores

$$m_i = ps_{\chi_i}(T_i), \qquad i = 1, 2.$$

Let $\tilde{\chi}$ be any extension of χ to T. The restriction of $\tilde{\chi}$ to each subtree T_i induces at least m_i changes, since m_i is minimal. Moreover, if the states assigned by $\tilde{\chi}$ to v_1 and v_2 differ, then at least one additional change must occur on one of the edges incident to ρ. Otherwise, no additional changes are required. Hence,

$$ps_\chi(T) \geq m_1 + m_2$$

if the states at v_1 and v_2 agree, or

$$ps_\chi(T) \geq m_1 + m_2 + 1$$

if the states do not agree.

It remains to show that for the extension produced by the algorithm the equalities hold. But this follows directly from the procedure: if the state sets on v_1 and v_2 intersect, no additional change is added. Otherwise, one change is added. This corresponds exactly to

$$m_1 + m_2,$$

or

$$m_1 + m_2 + 1$$

changes, respectively. Since no extension can achieve fewer changes than this, and the Fitch–Hartigan algorithm constructs an extension achieving this number, the algorithm computes the parsimony score $ps_\chi(T)$. ■

2.2.4 Computational cost

One of the most important properties that interests us about an algorithm is its computational cost. Let us now quantify the amount of work needed for

the performance of Fitch-Hartigan algorithm. We start with a single character on a set with n taxa and s possible states. One step of the algorithm on the internal vertex requires comparing the sets of states of its two children, assigning the proper set of states to that vertex, and possibly increasing the score. That amount of work grows linearly with s, therefore, we can write this as

$$Cs,$$

with C (unspecified) constant. A rooted binary tree has $n-1$ internal vertices. The work must be done on all of them, giving us

$$Cs(n-1).$$

In case of sequences of length k, that is, k characters, the process is performed for every character, resulting in the cost

$$Cs(n-1)k.$$

This can be expressed asymptotically as $O(skn)$.

The cost grows linearly in all variables, which is a very pleasant property. However, this is only the cost of computing the parsimony score of one tree. To obtain the most parsimonious tree, we would have to compute the score for each. This is discussed in more detail in the last section.

2.3 Comparison of T-Rex Generated Tree with Known Phylogeny

In this section, we compare two phylogenetic trees constructed for eight primate species. The first represents the widely accepted evolutionary relationships among these taxa, while the second was inferred with maximum parsimony based on the NR0B1 gene.

This gene is located on the reverse strand of chromosome X in position 30,304,206-30,309,390; therefore, its length is 5185 base pairs. It has two transcripts and encodes the protein DAX1, which is responsible for the development and function of endocrine glands, such as adrenal glands, hypothalamus, pituitary, and gonads. Mutations in NR0B1 lead to the underdevelopment of these organs, resulting in insufficient production of the corresponding hormones. Specific diseases caused by mutations include, for example, adrenal hypoplasia, which affects the adrenal glands, or hypogonadotropic hypogonadism, which causes insufficient production of sex hormones.

For the comparison, we consider the following species: Bonobo (*Pan paniscus*), Chimpanzee (*Pan troglodytes*), Gorilla (*Gorilla gorilla*), Human (*Homo sapiens*), Macaque (*Macaca mulatta*), Crab-eating macaque (*Macaca fascicularis*) (in the tree denoted as Macaque 2), Sumatran orangutan (*Pongo abelii*), Vervet-AGM (*Chlorocebus sabaeus*). Based on extensive genomic and morphological studies, the generally accepted phylogenetic relationships among these species are shown in Figure 5.

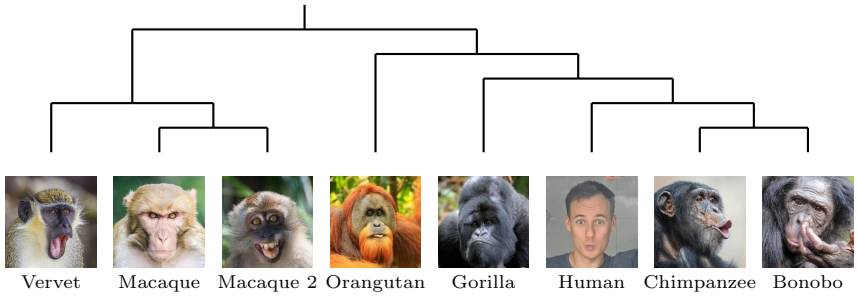

Figure 5: Known phylogenetic tree

The same phylogenetic tree is obtained also by using the software T-Rex with a Maximum Parsimony solver on the NR0B1 gene alignment, as shown in Figure 6. Note that the topology matches the known relationships, but T-Rex does not provide a rooted tree.

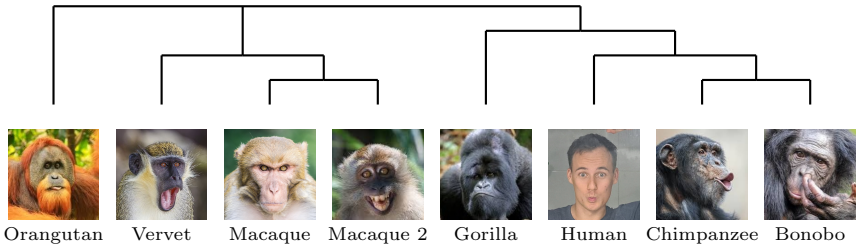

Figure 6: T-Rex generated tree

3 Implementation of the Fitch-Hartigan algorithm

3.1 When is a pattern informative?

First, we consider a DNA alignment of n species, which gives 4^n possible patterns (configurations of characters A, C, G, T at the leaves of the tree). We say that a pattern is non-informative if its parsimony score is the same for all tree topologies.

To see when a character is non-informative, we will look at 3 examples:

First, of course, if a character assigns the same state to all taxa, the character extends to that same constant character for all the internal vertices of any tree, and thus the parsimony score associated to it is obviously 0.

Another direct example is considering that a character assigns the same states to all taxa except to one, which yields just 1 change for any tree we consider.

A less direct cause is the following example: if a character assigns A to one taxon, G to another one and C to all the others, then any tree considered requires at least 2 changes, because we have 3 different states. Of course, if we join A and G by an interior node the tree will need 2 changes, and if we join A with a C, and G with another C, the required number of changes is again 2. Thus, independently of the chosen tree, this pattern will have a parsimony score of 2, and thus it doesn't provide any further information on which tree has the lowest parsimony score.

From these examples, we infer that a character is informative if and only if it assigns to the taxa *at least two* different states, each occurring at least twice. Let us formalize this reasoning by explaining why a character that does not satisfy this condition is non-informative:

Assume that χ is a character such that $|\chi^{-1}(i)| > 1$ for *at most one* state i. Denote by j a state for which $|\chi^{-1}(j)|$ is maximal, and let $l = |\chi(\mathbf{X})|$ be the number of distinct states. Then, for any tree T, we can extend χ to T by assigning the state j to each internal vertex. Thus, $ps_\chi(T) \leq l - 1$. Since, moreover, $ps_\chi(T) \geq l - 1$ for any character χ and any tree T, it follows that $ps_\chi(T) = l - 1$, which in particular does not depend on the tree T.

This type of character affects the parsimony score of all trees by inflating its value by the same amount. Thus, it does not affect the selection of the most parsimonious trees, which is why we say that it is a non-informative character.

We can therefore define an informative character as follows:

Definition. A character $\chi : \mathbf{X} \to S$ is *(parsimony)-informative* if

$$\left|\{\, i \in S : |\chi^{-1}(i)| > 1 \,\}\right| > 1.$$

We note that, before applying the Fitch–Hartigan parsimony algorithm, we can eliminate the non-informative characters in order to reduce the computational cost, and this will not affect the choice of the most parsimonious trees. We expect several non-informative characters to appear in DNA sequence alignments, since identical or nearly identical sites are needed to construct such alignments.

We also note that parsimony non-informative characters may be informative for other inference methods, as they contain relevant information about the resemblance between sequences.

Additionally, further reductions in computational effort are possible. In particular, we note that the specific labels of the states of a character do not matter, as long as the pattern of equalities and differences is preserved. For example, for a 5-taxon tree, an informative character assigning the states A, C, C, G, A yields the same parsimony score as one assigning G, A, A, T, G; for this reason, we work instead with patterns, in this case, $xyyzx$.

3.2 Number of non-informative patterns on a tree of n species where all sequences but one are in agreement

Now, let us consider n species of which all sequences but one are in agreement, that is, all taxa have the same state A, C, G or T except one, which has another state different from all the others (but also in the set $\{A, C, G, T\}$, of course). So, we have that all these patterns are non-informative, as there appear 2 different states, and one of them appears only once.

We can choose between 4 possibilities for the $n - 1$ sequences that are in agreement and we can choose between 3 possibilities for the different sequence, so we have $3 \cdot 4 = 12$ possible combinations of $n - 1$ sequences having the same state and the other one having a different state. Thus, if we take into consideration that we can choose any of the leaves of the graph to be the one with the different state, we can see that there are $12n$ possible different patterns.

3.3 How many patterns are non-informative due to 2 letters appearing once while all other letters agree?

Here, we consider that 2 letters appear once each, while all others agree. This means that these patterns are non-informative, since there are two different states that appear only once.

Regarding first the repeated letter, it can be chosen to be any state from the set $\{A, C, G, T\}$, so we have 4 possibilities. Once this letter is chosen, the "first" non-repeated letter can be chosen from 3 possibilities, and once this one is chosen, the final non-repeated letter can be chosen between 2 possibilities. Thus,

now we have $2 \cdot 3 \cdot 4 = 24$ possibilities for choosing the letters. Then, there are $\binom{n}{2}$ possibilities to choose which 2 leaves correspond to the non-repeated letters. Thus, the total number of patterns is

$$24\binom{n}{2}$$

3.4 How many patterns are non-informative due to 3 letters appearing once while all other letters agree?

Now, we consider that 3 letters appear once, and all others agree. Since there are 3 states appearing only once, this means that these patterns are non-informative.

Analogously to the two previous calculations, there are $4 \cdot 3 \cdot 2 \cdot 1 = 4! = 24$ possibilities for choosing which state A, C, G or T corresponds to the repeated letter and which states correspond to each of the non-repeated letters. Since there are $\binom{n}{3}$ possibilities to choose which 3 leaves correspond to the non-repeated letters, the number of patters is

$$24\binom{n}{3}$$

3.5 Efficient implementation to find MP score

An efficient implementation of the algorithm would be to first identify which are the informative sites, and just compute the score for the trees just for the informative sites. Then, we could add the score of the non-informative sites (which just needs to be computed once, and not each time for every tree) to the scores obtained for the informative sites for each different tree. This, however, doesn't solve the computational problem of the time needed for a big number of leaves, since there are 4^n different patters for n species, but we have seen in the previous sections that the number of non-informative patters can be just of the order of n, $\binom{n}{2}$ and $\binom{n}{3}$.

If we have the following alignment:

$$\begin{aligned}
a &: TGCGAGGCTAGACACGACATGGATTC \\
b &: AACCCTAGTATATGCGAGTATCTCTC \\
c &: AACATTGGTAGTTACCTGGTTCTCAG \\
d &: TACAGGACTCGTCTCTTCCCTGAGTT
\end{aligned}$$

To compute the MP score of the three possible unrooted trees, we will first have a look at each site independently, and see if it is informative or non-informative. If it is non-informative, we will just have to add the corresponding score of the site to all the trees.

We can see that the only positions which are informative are $\{1, 6, 7, 8, 12, 13, 17, 18, 22, 23\}$. So the three possible unrooted trees are $((a, b), (c, d))$, $((a, c), (b, d))$ and $((a, d), (b, c))$, which computation of the parsimony score corresponds to

the trees when using just the informative sites, i.e. $TGGCACACGA$ for sequence a, $ATAGATAGCT$ for sequence b, $ATGGTTTGCT$ for sequence c and $TGACTCTCGA$ for sequence d. We can see that in the Figures we have added a root, since the parsimony score doesn't depend on the position of the root.

Thus, for the informative sites of $((a,b),(c,d))$ the score is 18, for the informative sites of $((a,c),(b,d))$ the score is 19 and for the informative sites of $((a,d),(b,c))$ the score is 13. This is not the total parsimony score of the trees associated with the sequences a, b, c and d, since we still have to add the scores of the non-informative sites, but since the contribution of these sites to each one of the trees will be the same, we can conclude that the most parsimonious tree is going to be the third one, i.e. $((a,d),(b,c))$.

For the non-informative sites, since there are only 4 sequences, if there are two different states for a non-informative site the added score is going to be of 1, if there are three different states the added score is going to be 2 and if there are four states the added score is going to be of 3 (of course, if all states are the same the added score is going to be zero). Thus, adding all the scores of the non-informative sites yields a score of 23. This means that the parsimony score for the unrooted tree $((a,d),(b,c))$ is 36, for the unrooted tree $((a,b),(c,d))$ is 41 and for the unrooted tree $((a,c),(b,d))$ is 42.

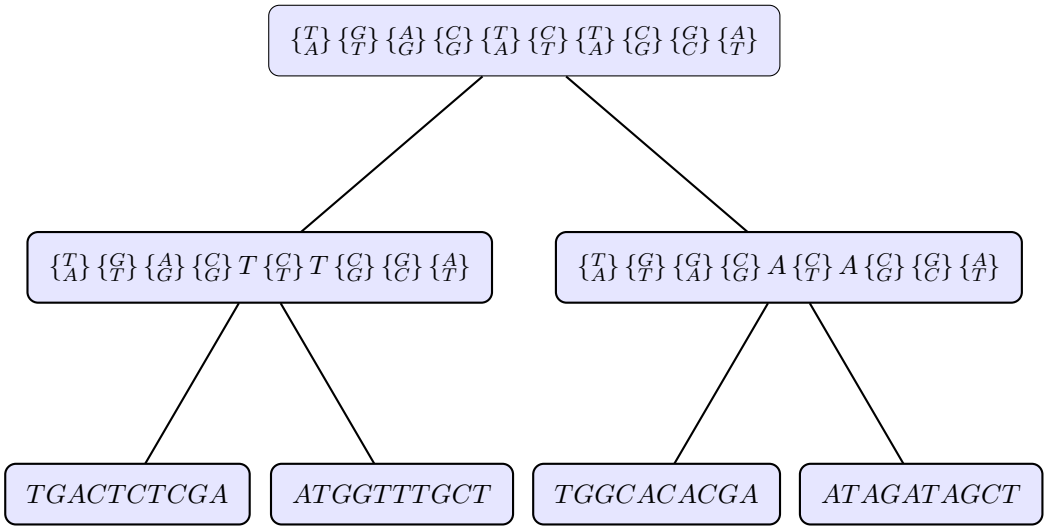

Figure 7: Computation of the parsimony score for the case $((a,b),(c,d))$.

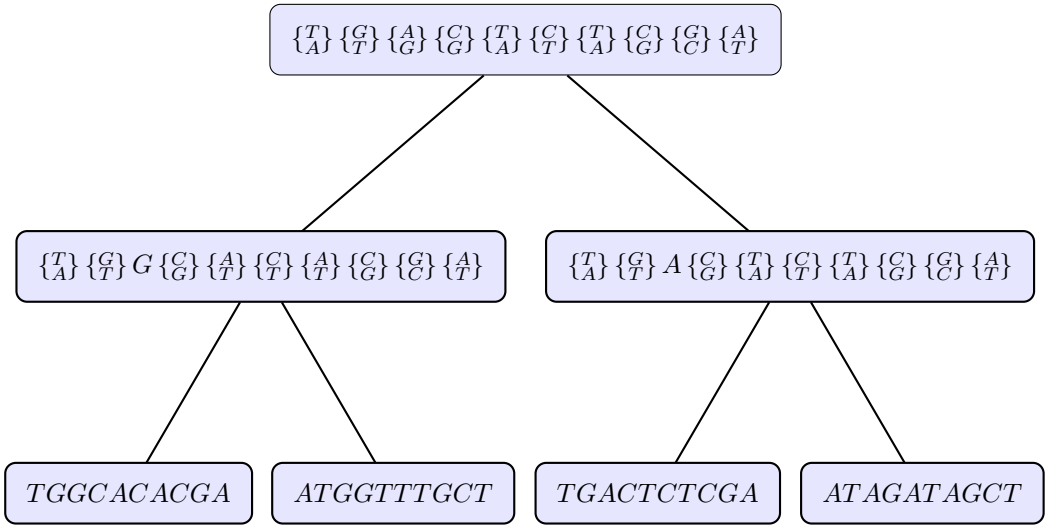

Figure 8: Computation of the parsimony score for the case $((a, c), (b, d))$.

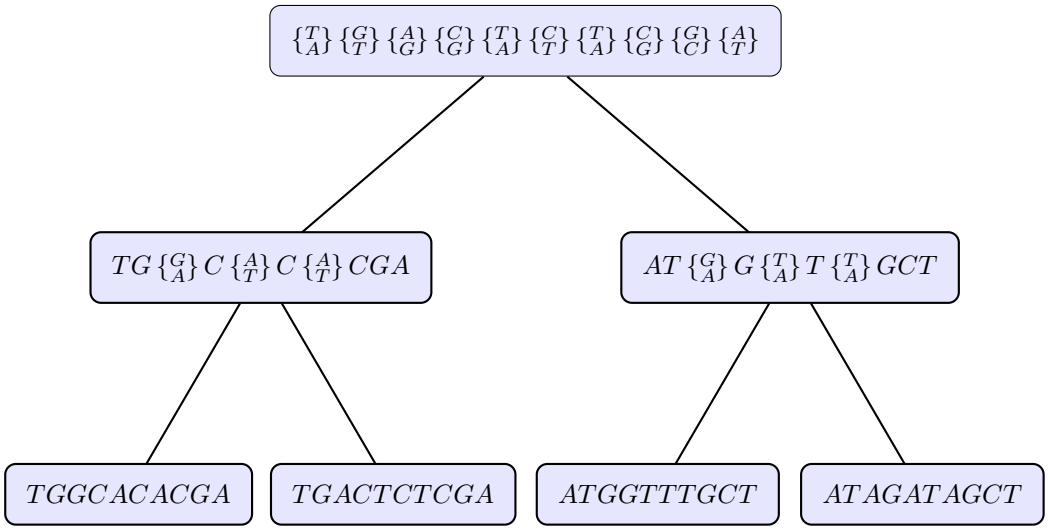

Figure 9: Computation of the parsimony score for the case $((a, d), (c, b))$.

4 Statistical Inconsistency of the Maximum Parsimony Method

Suppose that we could apply the MP method to a large amount of data, that is, to very long sequences. Would this lead to a more accurate phylogenetic tree inference than in the case where the amount of data is smaller?

More precisely, if parsimony is applied to longer and longer sequences generated according to a (known) phylogenetic tree model, do the inferred trees eventually stabilize on the correct one?

In this section, we explain why, for the MP method, the general answer to these questions is no, and how this outcome depends on the evolutionary model under consideration.

Consider an evolutionary model on a phylogenetic tree; that is, a tree T together with a set of random variables on the vertices of the tree taking values in the set of states, $\{X_v\}_{v \in V(T)}$, and a set of transition matrices on the edges of the tree, $\{M_e\}_{e \in E(T)}$.

To formalize the problem we are interested in, we may consider the following definition:

Definition. Let us consider an evolutionary model on a phylogenetic tree that depends on parameters s. Let $\hat{s}$ be the inferred value of the parameters from data, obtained from k independent trials using a given inference method.

We say that the inference method is *statistically consistent for the evolutionary model* if

$$\lim_{k \to \infty} \Pr(\|\hat{s} - s\| < \epsilon) = 1, \quad \forall \epsilon > 0.$$

If the inference method is statistically consistent for all possible evolutionary models, then we say that it is *statistically consistent*.

We note that when we refer to parameters of the evolutionary model, this can also include the tree topology.

Let us show that the *Maximum Parsimony method is statistically inconsistent*; that is, there *exist* phylogenetic trees that cannot be correctly inferred by the MP method as the amount of data increases.

Although the result has been generalized, here we follow the original argument introduced by Felsenstein, which provides a specific phylogenetic tree for which MP is statistically inconsistent.

Example on Statistical Inconsistency of the MP Method

Consider the phylogenetic tree with four leaves illustrated in Figure 11. Assume that data is generated on this tree taking values in a set of two possible states $S = \{0, 1\}$, that is, we are considering a 2-state model. Suppose that the root distribution and transition matrices of the tree are as follows:

$$\mathbf{p}_\rho = \left(\tfrac{1}{2}, \tfrac{1}{2}\right),$$

$$M_{(\rho,a)} = M_{(v,d)} = \begin{pmatrix} 1-q & q \\ q & 1-q \end{pmatrix},$$

$$M_{(\rho,b)} = M_{(\rho,v)} = M_{(v,c)} = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}, \qquad p, q \in (0, \tfrac{1}{2}).$$

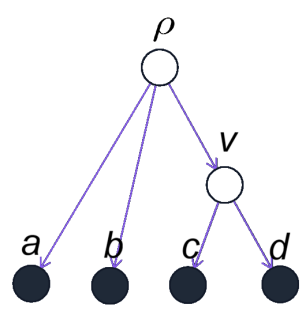

Figure 10: Phylogenetic tree under consideration. We note that the leaves' distribution is of the form of the quartet tree $T_1 : ab \mid cd$.

Now, suppose that data has been generated at the leaves of the tree, a, b, c, d, and we use this data to infer the underlying phylogenetic tree using the MP method.

First, we note that there exist three possible informative character patterns generated by the leaves a, b, c, d:

1. Pattern $a, b, c, d : xxyy$, which appears n_1 times.
2. Pattern $a, b, c, d : xyyx$, which appears n_2 times.
3. Pattern $a, b, c, d : xyxy$, which appears n_3 times.

For instance, if we generate the data:

$$\begin{array}{l} a: 0\ 0\ 1\ 1\ 1\ 0\ 1 \\ b: 0\ 0\ 1\ 1\ 0\ 1\ 1 \\ c: 0\ 1\ 1\ 0\ 1\ 1\ 0 \\ d: 0\ 1\ 0\ 0\ 0\ 0\ 0 \end{array}$$

then we count $n_1 = 3,\ n_2 = 1,\ n_3 = 1$.

Second, we note that from the generated data at the leaves, we can infer 3 possible quartet trees:

$$T_1 = ab \mid cd, \quad T_2 = ad \mid bc, \quad T_3 = ac \mid bd.$$

where T_1 is the real quartet tree generating the data.

We can compute the parsimony score of each quartet tree, in terms of n_i. For T_1, we note that each character pattern $xxyy$ will give a parsimony score of 1, while the other character patterns will produce scores of 2. Applying this reasoning also for the rest of quartet trees, we obtain:

$$\begin{aligned} ps(T_1) &= n_1 + 2n_2 + 2n_3 = (2n_1 + 2n_2 + 2n_3) - n_1, \\ ps(T_2) &= 2n_1 + n_2 + 2n_3 = (2n_1 + 2n_2 + 2n_3) - n_2, \\ ps(T_3) &= 2n_1 + 2n_2 + n_3 = (2n_1 + 2n_2 + 2n_3) - n_3. \end{aligned}$$

Now, since the tree generating the data is of the type T_1, we expect to infer this quartet tree; that is, we expect to obtain

$$ps(T_1) < ps(T_2), ps(T_3)$$

or, equivalently, to observe

$$n_1 > n_2, n_3.$$

In the limit of an infinite sequence of characters at each leaf of the tree, that is, when the sequence length $N \to \infty$, the ratios n_i/N tend to the probabilities p_i of obtaining each character pattern according to the phylogenetic tree:

$$n_i/N \to p_i.$$

Therefore, in the limit of infinite sequences, we expect to observe

$$p_1 > p_2, p_3,$$

which is a necessary condition for statistical consistency.

If we compute the probabilities $p_1 = p_{xxyy}$ and $p_2 = p_{xyyx}$ using the expression for the joint probability specific to this phylogenetic tree:

$$P_{ABCD} = \sum_{x,y\in\{0,1\}} P_\rho(x)\, M_{(\rho,a)}(x,A)\, M_{(\rho,b)}(x,B)\, M_{(\rho,v)}(x,y)\, M_{(v,c)}(y,C)\, M_{(v,d)}(y,D),$$

we obtain the expressions

$$p_1 = p_{xxyy} = (1-q)^2 p(1-p)^2 + 2q(1-q)p(1-p)^2 + q^2p^3,$$
$$p_2 = p_{xyyx} = (1-q)^2 p^2(1-p) + 2q(1-q)p^2(1-p) + q^2(1-p)^3.$$

Then, we have:

$$p_1 - p_2 = (1-2p)\big(p(1-p) - q^2\big),$$

which *does not* satisfy $p_1 > p_2$ *for all values of the parameters* p and q, implying statistical inconsistency of parsimony for the considered phylogenetic tree in those (p,q) values for which $p_1 \leq p_2$.

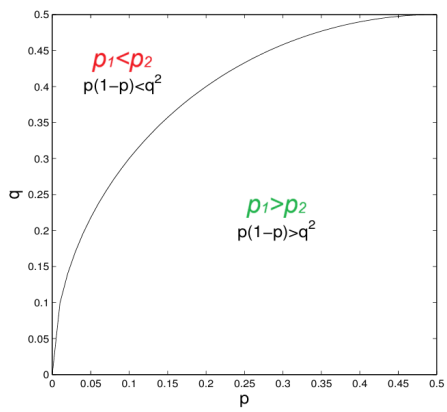

Figure 11: Regions on the (p,q) plane in which $p_1 > p_2$ and $p_1 < p_2$.

5 Limitations of Maximum Parsimony

5.1 Large Trees

Since the parsimony score can be used to compare the relative quality of different trees topologies, but does not provide a way to directly construct the optimal tree, the MP method requires searching through the space of possible tree topologies to find the one with the lowest parsimony score (large parsimony problem). This becomes computationally infeasible very quickly as the number of species increases. We will derive the number of possible tree topologies for n species and discuss the implications for using MP on large trees.

An unrooted binary tree with n leaves (species) has $2n-3$ edges. The number of distinct unrooted binary tree topologies $T(n)$ is given by the formula:

$$T(n) = (2n-5)!! = \frac{(2n-5)!}{2^{n-3}(n-3)!}$$

This formula can be derived recursively: We start with 3 species, which has only 1 possible tree topology. When we add a new species, it can be attached to any of the existing edges in the tree. Because all the leaves of the tree represent a different taxa, and therefore are distinguishable, every choice leads to a different tree topolgy. Since an unrooted binary tree with k leaves has $2k-3$ edges, adding the $(k+1)^{th}$ species increases the number of topologies by a factor of $2k-3$. Thus, the total number of topologies for n species is

$$1 \times 3 \times 5 \times \ldots \times (2n-5) = (2n-5)!!$$

The number of rooted binary tree topologies $T_{rooted}(n)$ is given by $T(n+1)$, since we can add an extra node which represents the root of the tree. Note that for the maximum parsimony method its enought to consider unrooted trees and choosing an arbitrary root, since the parsimony score does not depend on the position of the root. Already for $n = 20$ species, the number of possible unrooted tree topologies is approximately 2.22×10^{20}, making the maximum parsimony method infeasible.

5.2 Is parsimony the right assumption?

The maximum parsimony method is based on the assumption that evolution proceeds by the minimum number of changes. There are both arguments supporting and opposing this assumption.

Arguments supporting parsimony

- Simplicity: The principle of parsimony is elegant and computationally straightforward.
- Pragmatic Success: In many simple cases (including the phylogenetic tree of ape species mentioned above), parsimony has produced reasonable trees that match known phylogenies.
- Biological Intuition: Evolution by small steps of mutation and selection seems parsimonious.

Arguments against parsimony

- The idea of parsimony is a philosophical assumption, not an empirical law of biology.
- Statistical Inconsistency: Felsenstein (1978) proved that under certain conditions, maximum parsimony is statistically inconsistent, as discussed above.

- Biological Accuracy: There are several biological phenomena, that are not well modeled by parsimony, such as homoplasy (independent acquisition of the same character), different rates of evolution across lineages, and varying probabilities of different types of mutations. The process of gene mutations is not modeled.
- It does not provide branch lengths, which are important for understanding evolutionary distances.
- The method just compares tree topologies, but does not provide a way to generate new trees directly. This makes it computationally infeasible very quickly as the number of species increases.
- Empirical Counterexamples: There are cases where parsimony produces incorrect trees when the evolutionary model violates parsimony assumptions (e.g., fast-evolving lineages).
- Often is not unique - multiple trees can have the same minimum parsimony score, leading to ambiguity in the inferred phylogeny.
- Ignores non-informative sites, which can lead to loss of useful evolutionary information.

Overall, while maximum parsimony can be a useful tool for phylogenetic reconstruction, in many cases, especially when applied to taxa alignments, its replaced by different methods today (e.g., maximum likelihood, Bayesian inference).

5.3 Alternative methods for searching Trees topologies

Instead of exhaustively searching through all possible tree topologies, there are several heuristic methods that can be used to explore the tree space more efficiently. Two common operations used in these heuristics are Nearest Neighbor Interchange (NNI) and Subtree Prune and Regraft (SPR).